

Generation Interconnection System Impact Study Report

For Queue Project Q22-2113

Rhineland Paper 138 kV

January 16, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2113. The Interconnection Customer proposes a 298.8 MW solar photovoltaic facility interconnecting at the Rhineland Paper 138 kV substation in the Virginia area. The request was received on June 12, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 574 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,349,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Hartman Crk 138/1kV TX
Contingency: Lowmoor - Hinton Ap 345.0 kV L
Loading: 111.9% (Rating: 490 MVA)
DFAX: 18.3%
Pre-project loading: 100.7%
2. Facility: Hatman Creek 345 kV/1 kV XFMR
Contingency: Hinton Ap - Lowmoor 345.0 kV L
Loading: 107.7% (Rating: 509 MVA)
DFAX: 18.3%
Pre-project loading: 96.9%
3. Facility: Mount View - Hartman Creek 138.0 kV
Contingency: Lowmoor - Hinton Ap 345.0 kV
Loading: 104.6% (Rating: 431 MVA)
DFAX: 19.7%
Pre-project loading: 91.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hartman Crk 138/1 kV Transformer	Replace transformer with higher-rated unit	\$4,338,000	6% (\$263,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hartman Creek Substation 345/1 kV TX	Replace transformer with higher-rated unit	\$14,946,000	6% (\$906,000)
Hartman Crk - Mount View 138.0 kV	Reconductor line	\$1,095,000	16% (\$180,000)
	Total allocated to project		\$1,349,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.